Nonexistence of a Circulant CGW(35, 12; 6) and the Case CW(105, 36)

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Abstract

We give an exact computer-assisted proof that no circulant complex generalized weighing matrix CGW(35, 12; 6) exists. Reduction modulo the Eisenstein prime $1-\omega$ places the support of a hypothetical first row in one of two explicit ternary cyclic [35, 12] codes. Complete enumeration gives 420 possible supports, and an exact affine-orbit calculation shows that they form one orbit. On a canonical support, reduction modulo 2 yields a short contradiction from the four correlations at shifts 2, 4, 8, 10. The known three-fold contraction of a hypothetical CW(105, 36) would produce exactly such a sixth-root sequence, so its nonexistence follows. Python, standalone C++, result-blind SageMath and result-blind GAP implementations reproduce the finite classification by distinct routes; these computations are supporting validation, while the final characteristic-two obstruction is algebraic.

1 Statement and group-ring formulation

Let $\mu_6 = \langle \zeta_6 \rangle$, where $\zeta_6 = e^{\pi i/3} = 1 + \omega$ and $\omega^2 + \omega + 1 = 0$. A circulant CGW$(n, w; 6)$ is a circulant matrix with entries in $\{0\} \cup \mu_6$ whose product with its conjugate transpose is wI_n. Equivalently, its first row

$$u = (u_0, \dots, u_{n-1}) \in (\{0\} \cup \mu_6)^n$$

has Hamming weight w and periodic Hermitian correlations

$$C_u(t) = \sum_{q \in \mathbb{Z}_n} u_q \overline{u_{q+t}} = \begin{cases} w, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

Put

$$U(X) = \sum_{q=0}^{34} u_q X^q \quad \text{in} \quad \mathbb{Z}[\omega][X]/(X^{35} - 1),$$

and let # conjugate coefficients and send X to X^{-1}. Thus a circulant CGW(35, 12; 6) is exactly a weight-12 polynomial satisfying

$$UU^{\#} = 12. \tag{1}$$

No coordinate is prescribed to vanish.

2 The factor-orbit reduction

Lemma 1 (Factor-orbit lemma). *Let F be a field, let $M \in F[X]$ be square-free, and suppose # induces an involution on the irreducible factors of M. If*

$$vv^{\#} = 0 \pmod{M},$$

then every factor fixed by # divides v, and from every two-element orbit $\{f, f^{\#}\}$ at least one of $f, f^{\#}$ divides v.

Proof. Each irreducible factor f of the square-free polynomial M is prime. Therefore $f \mid vv^{\#}$ implies $f \mid v$ or $f \mid v^{\#}$. The second alternative is equivalent to $f^{\#} \mid v$. □

Reduce (1) modulo $\pi = 1 - \omega$. Since

$$\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_3$$

and every element of μ_6 maps to 1 or -1, the reduction V has exactly the same support as U. Moreover $12 = 0$ in $\mathbb{F}_3$, so

$$V(X)V(X^{-1}) = 0 \pmod{X^{35} - 1}. \tag{2}$$

The complete irreducible factorization over $\mathbb{F}_3$ is

$$X^{35} - 1 = (X-1)\Phi_5(X)\Phi_7(X)A(X)B(X),$$

where A and B are reciprocal irreducible polynomials of degree 12. In low-to-high coefficient notation,

$$A = [1, 2, 2, 1, 2, 1, 0, 1, 2, 0, 1, 0, 1],$$
$$B = [1, 0, 1, 0, 2, 1, 0, 1, 2, 1, 2, 2, 1].$$

All five displayed factors are distinct and irreducible, and their product is $X^{35} - 1$; this is therefore the complete square-free factorization. The first three factors are fixed by reciprocity, while $\{A, B\}$ is the sole two-element reciprocal orbit. Put $H = (X-1)\Phi_5\Phi_7$. Lemma 1 gives $H \mid V$ and either $A \mid V$ or $B \mid V$. Consequently the complete solution locus of (2) is the union of the two cyclic $[35, 12]$ codes generated by

$$G_A = (X-1)\Phi_5\Phi_7 A, \qquad G_B = (X-1)\Phi_5\Phi_7 B.$$

Conversely, a multiple of G_A has reciprocal divisible by G_B (and vice versa), so every word in either code satisfies (2). Their intersection is $\{0\}$: divisibility by both generators gives $X^{35} - 1 \mid V$. Since the reduction under consideration has support 12, it is nonzero and lies in exactly one of the two codes. Their coefficient vectors, again low-to-high, are

$$G_A = [2, 0, 1, 0, 1, 1, 0, 2, 0, 1, 2, 1, 1, 1, 0, 2, 0, 1, 1, 1, 2, 2, 1, 1],$$
$$G_B = [2, 2, 1, 1, 2, 2, 2, 0, 1, 0, 2, 2, 2, 1, 2, 0, 1, 0, 2, 2, 0, 2, 0, 1].$$

The standalone publication audit verifies the factor product, irreducibility, reciprocity, and both generator products exactly.

3 Complete ternary enumeration and its certificate

For each generator $G \in \{G_A, G_B\}$, form the 12 rows $G, XG, \ldots, X^{11}G$ in $\mathbb{F}_3^{35}$. Their leading coordinates are distinct, so they have rank 12. The enumeration visits every message in $\mathbb{F}_3^{12}$, multiplies it by this generator matrix, computes the Hamming weight, and retains the words of weight 12. Direct periodic correlation is checked on every retained word. The exact counts are

	$\langle G_A \rangle$	$\langle G_B \rangle$
messages enumerated	3^{12}	3^{12}
weight-12 codewords	420	420
distinct supports	210	210

Each support occurs twice, for the scalar pair $v, -v$, and the two support families are disjoint.

For an explicit coordinate anchor, the message

$$m = (1, 0, 0, 2, 0, 1, 0, 0, 0, 0, 0, 0)$$

in the G_B branch gives the word

$$v = (2, 2, 1, 2, 0, 0, 0, 2, 0, 0, 1, 0, 2, 0, 0, 0, 2, 0, 0, 0, 0,$$
$$1, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0).$$

It has zero periodic autocorrelation over $\mathbb{F}_3$ and support

$$S = \{0, 1, 2, 3, 7, 10, 12, 16, 21, 22, 26, 28\}. \tag{3}$$

Lemma 2 (Affine-orbit lemma)**.** *The 420 supports obtained above are exactly the orbit of S under*

$$q \longmapsto aq + b, \qquad a \in \mathbb{Z}_{35}^{\times}, \quad b \in \mathbb{Z}_{35}.$$

Proof. The exact enumeration serializes every support as a 35-bit integer, with bit q representing coefficient q. Independently, all $35\varphi(35) = 840$ affine maps are applied to S. The setwise stabilizer is

$$\{(a,b) = (1,0), (29,28)\},$$

so the orbit contains $840/2 = 420$ supports. Sorting both integer sets gives byte-for-byte equality with no duplicates or omissions. The complete ordered list accompanies this paper. □

The numeric decimal-lines certificate has SHA-256

`c067600256b37e077ee8a83889ec5e09347c397feea723466fb863ca2685f3d2`.

For completeness, affine relabelling preserves the Hermitian correlation condition directly. If $a \in \mathbb{Z}_{35}^{\times}$, $b \in \mathbb{Z}_{35}$, and $y_{aq+b} = \lambda x_q$ with $\lambda\overline{\lambda} = 1$, then, writing $C_h(x) = \sum_q x_q \overline{x_{q+h}}$, the change of variable $q \mapsto aq+b$ gives

$$C_{ah}(y) = C_h(x).$$

Multiplication by a permutes the nonzero shifts. Hence it is enough to exclude the single support S.

4 A four-shift obstruction over $\mathbb{F}_4$

Reduce U modulo 2. Then

$$\mathbb{Z}[\omega]/(2) \cong \mathbb{F}_4,$$

conjugation is $x \mapsto x^2 = x^{-1}$ on $\mathbb{F}_4^{\times}$, every active coefficient remains nonzero, and $12 = 0$. Write $x_j \in \mathbb{F}_4^{\times}$ for the value at $j \in S$ and abbreviate $x_a/x_b = x_a\overline{x_b}$. The support intersections at shifts $2, 4, 8, 10$ give

$$\begin{aligned}
C_2 &= \frac{x_0}{x_2} + \frac{x_1}{x_3} + \frac{x_{10}}{x_{12}} + \frac{x_{26}}{x_{28}},\\
C_4 &= \frac{x_3}{x_7} + \frac{x_{12}}{x_{16}} + \frac{x_{22}}{x_{26}},\\
C_8 &= \frac{x_2}{x_{10}} + \frac{x_{28}}{x_1},\\
C_{10} &= \frac{x_0}{x_{10}} + \frac{x_2}{x_{12}} + \frac{x_{12}}{x_{22}} + \frac{x_{16}}{x_{26}} + \frac{x_{26}}{x_1} + \frac{x_{28}}{x_3}.
\end{aligned}$$

Suppose all four vanish. Because the field has characteristic two, $C_8 = 0$ says its two nonzero summands are equal. Put

$$p = \frac{x_2}{x_{10}} = \frac{x_{28}}{x_1}.$$

Multiplication by p gives

$$pC_2 = \frac{x_0}{x_{10}} + \frac{x_{28}}{x_3} + \frac{x_2}{x_{12}} + \frac{x_{26}}{x_1},$$

the sum of four terms of C_{10}. Hence $C_2 = C_{10} = 0$ forces

$$\frac{x_{12}}{x_{22}} = \frac{x_{16}}{x_{26}}, \qquad \text{and therefore} \qquad \frac{x_{12}}{x_{16}} = \frac{x_{22}}{x_{26}}.$$

The last two summands of C_4 are equal and cancel, leaving

$$C_4 = \frac{x_3}{x_7} \neq 0,$$

a contradiction.

Theorem 1. *No circulant* CGW$(35, 12; 6)$ *exists.*

Proof. The characteristic-three reduction of a hypothetical matrix has one of the 420 enumerated supports. Lemma 2 moves it to S, where the characteristic-two reduction contradicts the four displayed correlations. □

5 The $\mathrm{CW}(105, 36)$ corollary

We import exactly one multiplier result. Theorem 4.1 of Arasu–Gordon–Zhang [1], attributed there to McFarland, states that if $M \in \mathrm{ICW}_d(m, k)$, $\gcd(m, k) = 1$, and an integer t is congruent modulo m to a power of each prime divisor of k, then t is a multiplier of M. This theorem is an external dependency, not a computation in the present work.

For a hypothetical $\mathrm{CW}(105, 36)$, write $A(X) = \sum_{j=0}^{104} a_j X^j$. Its three-fold contraction is defined coefficientwise by

$$b_q = \sum_{r=0}^{2} a_{q+35r}, \qquad q \in \mathbb{Z}_{35}.$$

Then $B(X) = \sum_{q=0}^{34} b_q X^q$ is an $\mathrm{ICW}_3(35, 36)$, as in the contraction preceding Lemma 2.5 of the cited paper. Every hypothesis of Theorem 4.1 is explicit:

$$\gcd(35, 36) = 1, \qquad 36 = 2^2\, 3^2, \qquad 4 \equiv 2^2 \pmod{35}, \qquad 4 \equiv 3^{10} \pmod{35}.$$

Thus 4 is a multiplier of the contraction. Since $\gcd(4 - 1, 35) = 1$, a translate is fixed by 4. The complete finite enumeration of these fixed integer rows, independently reproduced by two Python enumerators and a standalone C++ sweep, leaves exactly

$$\begin{aligned} B_1(X) &= -3 + 3X^5 + 3X^{10} + 3X^{20}, \\ B_2(X) &= -3 + 3X^{15} + 3X^{25} + 3X^{30}. \end{aligned}$$

We treat both rows directly; their affine equivalence is not used below.

Use the CRT coordinates $i = 3q + 35r$, with $q \in \mathbb{Z}_{35}$ and $r \in \mathbb{Z}_3$. The two rows become, respectively,

$$\begin{aligned} \widetilde{B}_1(X) &= -3 + 3X^{15} + 3X^{25} + 3X^{30}, \\ \widetilde{B}_2(X) &= -3 + 3X^5 + 3X^{10} + 3X^{20}. \end{aligned}$$

For either row, the unique column of sum -3 is $(-1, -1, -1)$ and the three columns of sum 3 are $(1, 1, 1)$. A column of sum zero is either zero or one of the six permutations of $(1, -1, 0)$. The four constant columns contribute 12 to the weight; the remaining weight is 24, and each active zero-sum column contributes 2. Hence either contraction row forces exactly twelve active columns.

For a column (a_0, a_1, a_2) take its nontrivial fibre character

$$D_q = a_0 + a_1\omega + a_2\omega^2.$$

The six possible active-column patterns are

e	0	1	2	3	4	5
(a_0, a_1, a_2)	$(1, -1, 0)$	$(1, 0, -1)$	$(0, 1, -1)$	$(-1, 1, 0)$	$(-1, 0, 1)$	$(0, -1, 1)$

and satisfy $D_q = (1 - \omega)\zeta_6^e$. Thus $D = (1 - \omega)U$ for a length-35 sequence U with twelve nonzero sixth roots; constant and zero columns have character value zero.

Here is the projection coefficient by coefficient. Put

$$R(t, s) = \sum_{q \in \mathbb{Z}_{35}} \sum_{r \in \mathbb{Z}_3} a_{q,r} a_{q+t,r+s}.$$

The CRT map is a group isomorphism, so the $\mathrm{CW}(105, 36)$ equations say $R(0, 0) = 36$ and $R(t, s) = 0$ for every other pair. Substituting $u = r + s$ gives

$$\begin{aligned} \sum_q D_q \overline{D_{q+t}} &= \sum_{q,r,u} a_{q,r} a_{q+t,u} \omega^{r-u} \\ &= \sum_{s=0}^{2} \omega^{-s} R(t, s) = \begin{cases} 36, & t = 0, \\ 0, & t \neq 0. \end{cases} \end{aligned}$$

Since $D = (1 - \omega)U$ and $(1 - \omega)(1 - \overline{\omega}) = 3$, the preceding identity gives

$$DD^{\#} = 3UU^{\#} = 36.$$

The group ring $\mathbb{Z}[\omega][C_{35}]$ is free as an abelian group, so multiplication by 3 is injective and the factor cancels coefficientwise. Hence $UU^{\#} = 12$. This would be a circulant $\mathrm{CGW}(35, 12; 6)$, contrary to Theorem 1.

Theorem 2. *No circulant weighing matrix* $\mathrm{CW}(105, 36)$ *exists.*

6 Reproducibility and evidence boundary

The proof-bearing computation is the complete enumeration of two ternary $[35, 12]$ codes and the equality of their 420 supports with one affine orbit. Python and standalone C++ implementations regenerate the complete ordered certificate independently. SageMath 10.9 and GAP 4.12.1 implementations reproduce the classification by distinct CAS routes. These are independent implementations within the present project, not independent human authorship.

A separate C++ checker ignores the affine reduction and checks all

$$420 \cdot 3^{11} = 74{,}401{,}740$$

global-phase-normalized $\mathbb{F}_4$ assignments, finding no solution. This is redundant validation, not a logical dependency of the four-shift argument. An independent standard-library audit checks the factor branches, explicit word, support certificate, affine stabilizer, four correlation term lists, fibre table, contraction convention, and the publication manifest.

All theorem decisions use exact integer or finite-field arithmetic; no incomplete or non-exact computation is used. The imported contraction theorem and the apparently new support-preserving two-characteristic obstruction both remain subject to specialist review. Related published work and the maintained matrix repository are cited in [3, 2]; no priority claim is made beyond those sources.

Declaration of generative AI and AI-assisted technologies

During the development of this work, the author used OpenAI GPT-5.6 Sol, accessed through ChatGPT and Codex, as a substantive interactive research and software-development assistant. The model assisted in exploring candidate algebraic reductions, formulating and stress-testing proof steps, generating and reviewing exact verification code, designing adversarial checks, and organizing and editing the manuscript. Its outputs were treated as proposals, not as mathematical evidence or authoritative sources. The author reviewed all retained arguments, code, computations, and citations; they were accepted only after exact checks, cross-implementation comparison, or direct source verification. The author takes full responsibility for the mathematical claims, software, citations, and final manuscript.

References

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